CVPR #10

Pseudo-Hierarchies Improve Optimization Hyperparameter Robustness in Fine-Grained Image Recognition

Anonymous CVPR submission

Paper ID 10

Abstract

The backbone is an important component of a fine-grained image recognition system. While transformers have made significant progress in academic benchmarks, real-world deployments continue to rely on traditional CNN backbones. Our study reveals a key reason: many modern backbones exhibit pronounced sensitivity to hyperparameter settings, undermining their robustness in practical scenarios or increasing hyperparameter search costs. Hierarchical labels can mitigate this issue by regularizing the optimization process, but such annotations are rarely available—especially in specialized domains like leaf cultivar identification. Therefore, we introduce Hungarian Pseudo-Hierarchies (HPH), a training framework designed to inject hierarchical structure without requiring explicit label taxonomies. HPH begins by extracting features using a pretrained backbone, then aggregates each fine-grained class into compact clusters. A Hungarian bipartite matching module assigns these clusters to a reduced set of pseudo-classes, yielding a set of pseudo-hierarchical targets for all training samples. Finally, a fine-tuning stage leverages any standard hierarchy-aware training strategy to incorporate this synthetic structure. Across diverse settings—21 backbones, 4 fine-grained datasets, and a sweep of commonly tuned optimization hyperparameters—incorporating hierarchical biases enhances robustness to hyperparameter choices by increasing the average accuracy and reducing the standard deviation, and therefore reducing hyperparameter searches costs.

1. Introduction

Fine-grained image recognition tasks—such as distinguishing bird species or car models—require recognizing subtle visual differences between closely related categories. In these settings, the choice of backbone architecture is a central design decision [19, 21, 41]. Prior work has often attributed the strong performance of convolutional neural networks (CNNs) [14] on datasets such as FGVC-Aircraft [26] or Stanford Cars [23] to architectural properties like convolutional inductive biases that favor localized pattern recognition [41]. This view is also reflected in practice: despite the dominance of Vision Transformers in academic benchmarks, CNNs remain widely used in real-world workflows, appearing in over 70% of competition solutions according to an analysis of Kaggle competitions [2–4]. However, our experiments in Section 5 show that Vision Transformers [12] can achieve competitive performance on these fine-grained datasets. ViTs' underperformance can be due to strong sensitivity to optimization hyperparameters—such as learning rate, weight decay, and optimizer choice—which can lead to suboptimal results under commonly used defaults and create the misleading impression that a model family is poorly suited to the task. Furthermore, hyperparameter searches are costly and contribute significantly to the carbon footprint of deep learning pipelines [25, 31], making hyperparameter robustness a necessity for efficient training of models.

To address this issue, we incorporate hierarchical training objectives aligned with the structured label spaces common in fine-grained image recognition such as biological taxonomy. Instead of relying solely on flat cross-entropy at the finest level, hierarchical losses introduce supervision across multiple semantic levels (e.g., genus, family, species). This multi-level signal provides denser and more stable gradients, effectively regularizing optimization by guiding models through a coarse-to-fine learning process. As a result, the training dynamics become less sensitive to specific hyperparameter choices, reducing variance across configurations and enabling a fairer comparison of backbone architectures.

Our study offers three main contributions:

1. We study 21 pretrained backbones (11 CNNs, 10 ViTs) across four fine-grained image recognition datasets. Our results show that many backbones are sensitive to optimization hyperparameter choices.
2. We evaluate hierarchical supervision across backbones

and datasets. Hierarchy consistently improves optimization hyperparameter robustness, as measured by the coefficient of variations (CV) of the accuracy, by reducing CV up to **75%** compared to FGIR baselines.

3. We propose Hungarian Pseudo-Hierarchy (HPH), a novel pseudo-hierarchy construction strategy that alleviates the cost of manual annotation and enables incorporating hierarchical inductive biases into datasets without an explicit hierarchy. Experiments show that pseudo-hierarchies can improve robustness, reducing CV by up to **50%**.

2. Related Works

2.1. Fine-Grained Image Recognition (FGIR)

Early FGIR methods relied on part annotations or bounding boxes to guide localization [42]. While effective, these methods are difficult to scale due to the cost of annotations. Later works leveraged image-level labels with attention mechanisms, such as MA-CNN [43], which learned part-aware features without extra supervision. However, these methods often employ multi-stage pipelines involving zooming-in and cropping discriminative areas, which significantly increases the cost.

Kornblith et al. [21] and follow-up work [13, 19] demonstrated that the backbone can make as much of a difference as a specialized discriminative part selection method. Ye et al. [41] investigated backbones for weakly supervised FGIR and concluded that CNN backbones are more suitable for machine-like tasks such as aircraft and cars classification. But our results on Section 5 show that ViTs can be competitive to CNNs even in this tasks; the reason why ViTs may be perceived inferior in these tasks is due to sensitivity to optimization hyperparameters.

2.2. Hierarchical Fine-Grained Image Recognition

Hierarchical FGIR methods can address hyperparameter sensitivity by introducing inductive biases that act as regularization in the form of multi-level prediction heads and hierarchy-aware losses. One of the first works, Multi-Granularity Descriptor (MGD) [36], used weakly supervised taxonomic descriptors. This was followed by methods such as, Hierarchical Semantic Embedding (HSE) [9] and granularity-specific classifiers [7], which introduced hierarchy-aware loss functions and multi-level prediction heads. Although effective, these methods still struggle with class ambiguity within coarse groups and rely on a narrow set of backbone architectures.

2.3. Hierarchically Consistent Learning

Hierarchically consistent learning enforces compatibility between predictions at different hierarchy levels through architectures or loss functions. Early work such as Hierarchical Semantic Embedding (HSE) by Chen et al. [9] sequentially predicts category levels and uses higher-level predictions as soft constraints to regularize lower-level output. Hierarchical Residual Network (HRN) [8] proposed injecting coarse-level features into finer-level through residual connections, allowing parent-category information to guide fine-grained recognition.

Later approaches extend consistency enforcement through different mechanisms, such as Semantics-Aware Hierarchical Consensus (SAHC) [38] aligns predictions from multiple hierarchy levels through a consensus distribution optimized via divergence minimization, improving robustness under bad hierarchies. Free-Grained Hierarchical Recognition [28] addresses mixed-granularity supervision using vision–language models to propagate semantic information across hierarchy levels.

Collectively, these works demonstrate that explicitly enforcing hierarchical agreement can increase performance and robustness. Our method is complementary to these approaches; our method creates pseudo-hierarchies which can be integrated with existing hierarchical training frameworks, enabling incorporating hierarchical inductive biases into datasets without predefined hierarchies.

2.4. Pseudo-Hierarchies in Image Classification

Pseudo-hierarchies are widely used across different image classification tasks, from unsupervised person re-identification (re-ID) to HFGIR. These methods construct hierarchical structures from data representations to improve learning efficiency. In unsupervised re-ID, Online Hierarchical Cluster Dynamics [44] forms a bottom-up hierarchy from feature embeddings that evolves during training. Logits-to-Hierarchy (L2H) [27] derives hierarchical class structures by clustering classifier logits to recover semantic taxonomies without retraining. Deep Taxonomic Networks [37] learn hierarchical organization by imposing a tree-structured Gaussian mixture prior over latent representations, while pseudo-fine labels can also be constructed under coarse supervision using feature clustering [17].

Existing methods follow a coarse-to-fine (top-down) clustering approach, whereas our method constructs pseudo-hierarchies from feature representations of pretrained models and existing fine labels, merging categories into coarser groups (bottom-up) to facilitate optimization of backbones under fine-grained conditions.

3. Learning with Hungarian Pseudo-Hierarchies

Our approach consists of three stages. First, feature representations are extracted from pretrained backbones. Second, Hungarian Pseudo-Hierarchies (HPH) are constructed by clustering fine-grained classes into coarse classes based

on their feature representations. Finally, the resulting hierarchy is used within a hierarchical module for training and prediction. The following subsections describe feature extraction, pseudo-hierarchy construction, and hierarchical module.

3.1. Feature Representation Extraction

Features are extracted from pretrained backbones. For CNN-based models (ResNet50 variants), features are extracted from the last BatchNorm layer of the final residual block, while for Transformer-based models (ViT-B variants) features are obtained from the last LayerNorm of the encoder. Feature vectors are pooled with adaptive average pooling to ensure a consistent output shape.

Let x be an input image and $\phi(\cdot)$ the feature extractor. The extracted representation for a single image is

$$\mathbf{z} = \phi(x) \in \mathbb{R}^C,$$

where C is the number of feature channels. Collecting features across all images in a dataset yields a feature matrix

$$Z \in \mathbb{R}^{S \times C},$$

where S is the number of samples. These features preserve semantic information at the fine-grained class level and provide a foundation for clustering-based hierarchy construction.

3.2. Pseudo-Hierarchy Construction

To alleviate the cost of manually annotating hierarchical labels, we propose **Hungarian Pseudo-Hierarchies (HPH)**, a method that constructs pseudo-hierarchies from fine-grained labels by formulating coarse-level grouping as a bipartite matching problem. The resulting coarse labels are mapped to fine classes, producing a hierarchy that can be integrated with any hierarchical FGIR module.

Given feature representations extracted from a pretrained backbone, each fine class f_i is represented by its centroid:

$$\mathbf{c}_i = \frac{1}{|k_i|} \sum_{x \in k_i} \phi(x),$$

where k_i denotes the set of samples belonging to fine class i, and $\phi(x) \in \mathbb{R}^d$ is the d-dimensional feature embedding (channel representation) extracted from sample x. Each centroid $\mathbf{c}_i$ lies in the same feature space as the backbone feature representations.

We then compute the pairwise Euclidean distance between centroids to form a cost matrix:

$$C_{ij} = \sqrt{\sum_{m=1}^{d} (c_{i,m} - c_{j,m})^2}, \quad C \in \mathbb{R}^{N \times N},$$

where N is the number of fine classes.

Since the desired number of coarse classes L is smaller than N, we pad the cost matrix with the maximum distance for the extra columns. The Hungarian algorithm [24] is then applied to find the optimal one-to-one assignments between fine and coarse classes. Fine classes assigned to padded columns are reassigned to the nearest valid coarse cluster based on centroid distance. This procedure generates a 2-level pseudo-hierarchy for a dataset.

3.3. Hierarchical Classification Module

Given a hierarchy constructed either from manual annotation or the proposed pseudo-hierarchy generation process, prediction is performed using a hierarchical multi-granularity classification framework following Chang et al. [7]. In this setting, each image is associated with labels ordered from coarse to fine granularity. The objective is a multi-task learning problem in which classifiers at different semantic levels are learned jointly.

Separate classification heads are assigned to each hierarchy level, enabling independent prediction of coarse- and fine-level labels. This design disentangles the decision spaces across different hierarchical levels, allowing each classifier to specialize in recognizing patterns relevant to its semantic level.

Let the backbone network produce a feature representation $\mathbf{z}$ for an input image x. For a hierarchy with K levels, the shared feature is forwarded to K classification heads $g_k(\cdot)_{k=1}^{K}$, where each head predicts a probability distribution over the label space of its corresponding level with the formula below:

$$\mathbf{p}^{(k)} = g_k(\mathbf{z}), \quad k = 1, \ldots, K,$$

Training is performed using $\mathcal{L}_{\text{CE}}$ computed independently at every hierarchy level. The overall hierarchical loss is the sum of per-level losses:

$$\mathcal{L} = \sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)}$$

Each classifier therefore contributes to the overall optimization while sharing the same backbone representation.

The aggregated loss supervises predictions across all levels in the hierarchy, encouraging the learned representation to remain consistent with the hierarchical structure. During inference, the model outputs one prediction per level. The collection of predictions across levels forms the final hierarchical output, completing the classification process.

4. Experiment Setup

We evaluate our approach on four datasets: **FGVC Aircraft** [26], **CUB-200-2011**, **Soylocal** [35], and **Stanford Cars** [22]. Real hierarchies are constructed from Wikipedia class taxonomies following Chang et al. [7].

Our experiments consider 21 backbones spanning two types, 10 Transformer (ViT-B) and 11 CNN (ResNet50), covering a broad range of training methods, including fully supervised (FSL), semi-supervised (SemiSL), and self-supervised learning (SSL). Here are the exact models used:

- **ViT-B models**:
  - **FSL**: ViT[12], DeiT[33], DeiT3 (IN1K, IN21K)[33], MIIL [29]
  - **SSL (Discriminative)**: MoCoV3[10], DINO[6], CLIP-LAION5B[11], SigLIP-v2[34]
  - **SSL (Generative)**: MAE[15]
- **ResNet50 models**:
  - **FSL**: Torchv1[14], Torchv2[14], Gluon[16], A1[39], MIIL[29]
  - **SemiSL**: IG1B[40], YFCC100M[40]
  - **SSL (Discriminative)**: MoCoV3[10], SupCon[20], SwAV[5]
  - **SSL (Generative)**: SparK[32]

To assess model effectiveness in hierarchical FGIR, we measure several evaluation metrics measuring individual level accuracy and robustness:

- **Finest-level accuracy**. We report the Top-1 classification accuracy at the finest level of the hierarchy (L1). While the hierarchical structure contains up to three levels (L3, L2, and L1, from coarsest to finest), the coarse labels in the pseudo-hierarchy are automatically generated and do not correspond to ground-truth semantic labels. Therefore, coarse-level accuracies are not meaningful for evaluation. As a result, we focus our analysis on the finest-level accuracy, which reflects the true classification performance on the original dataset labels.
- **Coefficient of Variation (CV)**. Coefficient of Variation [1] balances accuracy and robustness by measuring the ratio of standard deviation to mean accuracy across multiple runs. Coefficient of Variation is defined as:

$$CV = \frac{\sigma}{\mu} \times 100\%$$

where μ denotes the mean accuracy and σ the standard deviation across evaluation runs. Lower Coefficient of Variation indicates a model achieves both high mean accuracy and low deviation, desirable characteristics in practical deployments.

We fine-tune our models for 200 epochs with a batch size of 8. To study the impact of hyperparameters, we use two optimizers (SGD and AdamW) across 10 learning rates, SGD (0.3–0.003) with weight decay 0, and AdamW (0.005–0.000005) with weight decay 0.05. We employ cosine learning rate scheduler and mixed precision. During training, we employ standard data augmentation (resize to 550×550, random crop to 448×448, horizontal flip). Center cropping is used during inference.

We control the granularity of coarse-level pseudo-hierarchies using a **cluster ratio**, defined as the number of coarse labels relative to the number of fine-grained labels. We construct 2-level pseudo-hierarchies with three cluster ratios: 25%, 50%, and 70%. These hierarchies are used to evaluate how the number of coarse-level labels affects hierarchical FGIR performance.

5. Results

5.1. Impact of Backbone, Pretraining and Hyperparameter on FGIR

5.1.1. Backbone Choice Drives Performance

Fig. 1 and Fig. 2 highlight how backbone selection alone can lead to vastly different outcomes. Among ResNets, MoCoV3 performs best (94% wAP on Aircraft), while the worst performing one, Spark obtains 92% wAP. ViT models show larger performance variance: SigLIP-v2 excels on Aircraft (94%) and Cars (95%), while original ViT excels CUB (91%). In contrast, DINO performs worst with 87% on Aircraft and 89% on Cars. These findings show that backbone selection, especially for ViTs, can lead to **state-of-the-art performance or degraded results**.

5.1.2. ResNet models do better on machines, but ViT models do better on animals

In Fig. 1 and Fig. 2 **ResNet models** generally **perform better** on **Aircraft and Cars**, while **ViTs do better** on **CUB**. This aligns with Ye et al.[41], which suggests that **objects with more dynamic shapes** (e.g. animals) **benefit** more from the **self-attention mechanism** in transformers. In contrast, **objects with more static shapes** (e.g. machines) **benefit** from the **inductive biases** in CNNs. Also, ImageNet contains a substantial number of bird-related images, which may have contributed to a performance boost for ViT models on the CUB dataset.

However, we observe **an exception**: **SigLIPv2**, a ViT model, achieves **better performance** on **Aircraft and Cars** than the **best ResNet model**, challenging this observation. To explain this, we employ **Mean Silhouette Coefficient (MSC)** [30], which quantifies the quality of feature clustering based on inter-class separation[18] and intra-class variance[18]. Higher values indicate well-separated and compact class clusters, while lower values imply more ambiguous features. We plot the Top-2 and Worst-2 wAP vs MSC for clarity, as observed in 3, SigLIP obtains lower MSC compared to other ViTs, suggesting tighter and less distinct feature representations. We also observed that ResNets obtains lower MSC than ViTs, while SigLiP obtains lower values than ResNets. This indicates that **performance stems from learned representations during pre-**

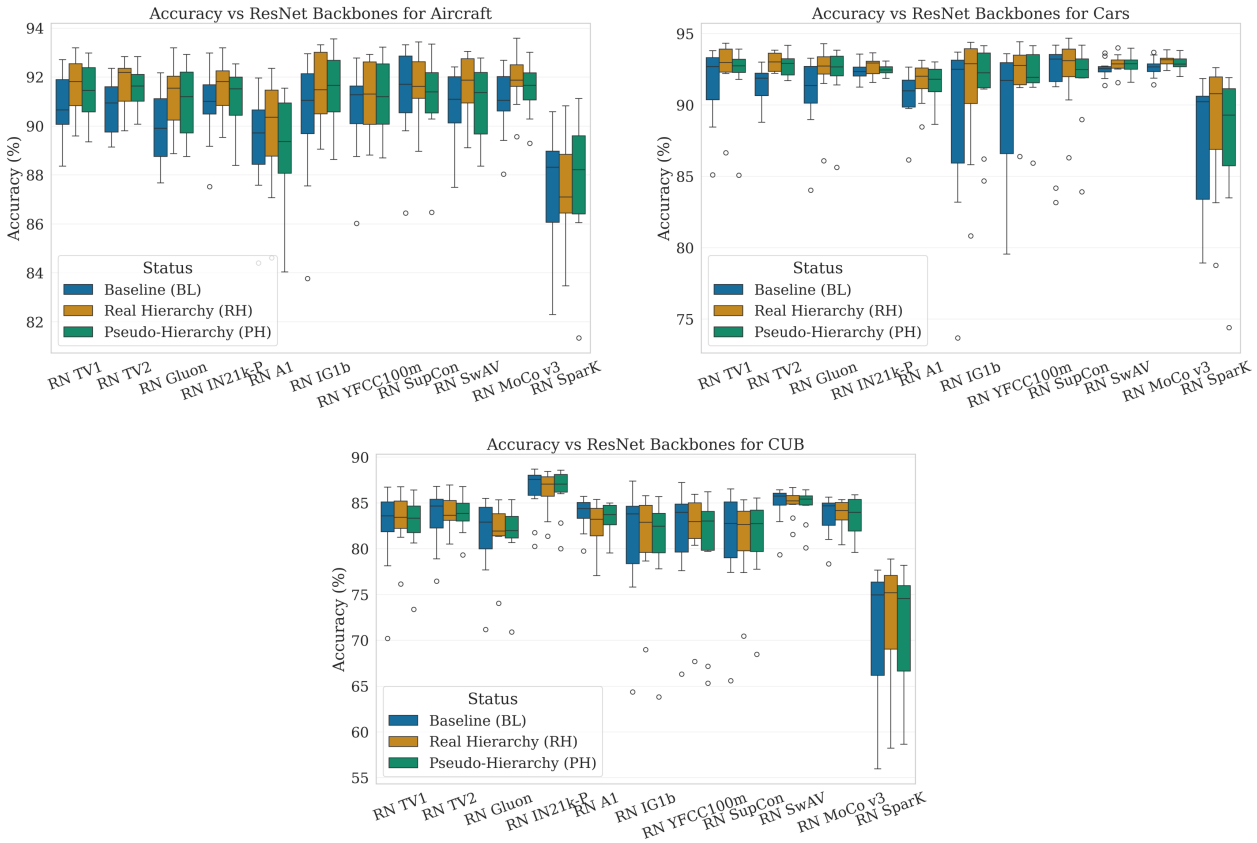


Figure 1. Performance of different ResNet models across 3 settings: **Baselines** (Without hierarchy), **Real Hierarchies** and **Pseudo-Hierarchy** and 3 datasets. Pseudo-hierarchy for Aircraft and Cars use 50% cluster ratio, while CUB uses 70%.

training rather than only architectural biases.

5.1.3. ResNet models are more robust than ViT models

In Fig. 1 and Fig. 2, under **fine-tuned settings**, **ResNet models** pretrained with **discriminative objectives** (e.g., MoCoV3, SwAV) achieve the highest mean accuracy and lowest deviation. For **ViT models**, **fully supervised models** such as original ViT, DeiT, and DeiT3 also show strong fine-tuned performance with low deviation. In general, **ResNet models are more robust** than ViTs across pretraining and hyperparameter, as supported by consistently lower CVs in both figures. Among them, MoCoV3, a discriminative model, achieves the best CV across different settings. These results show discriminative ResNet models as a strong baseline for future hierarchical FGIR research.

5.2. Impact of introducing Hierarchy on FGIR

5.2.1. Introducing hierarchy improves robustness

Fig. 1 and Fig. 2 show that hierarchical supervision consistently improves robustness across architectures, as indicated by lower CV compared to baseline. In ResNet models, hierarchy can reduce CV by almost half. For example, ResNet IG1b on Aircraft drops CV from 3.079 to 1.729, which is almost half. This effect is more pronounced for ViT models, which are more sensitive to hyperparameter choices. ViT base on Aircraft drops CV by nearly fourfold from 24.997 to 6.250.

5.2.2. Hierarchy is more effective on machine-centric datasets

The impact of real hierarchical supervision varies across datasets, with larger CV reductions observed on Aircraft and Cars as shown in Fig. 1 and Fig. 2. For example, ResNet IG1b sees CV decrease from 3.079 to 1.729 on Aircraft and from 6.691 to 0.698 on Cars, whereas on CUB the reduction is smaller, from 7.035 to 6.191. This trend may reflect dataset-specific characteristics discussed in our backbone analysis 5.1.2, where more static shapes in Aircraft and Cars compared to more dynamic shapes in CUB, could influence the effectiveness of hierarchy in improving robustness.

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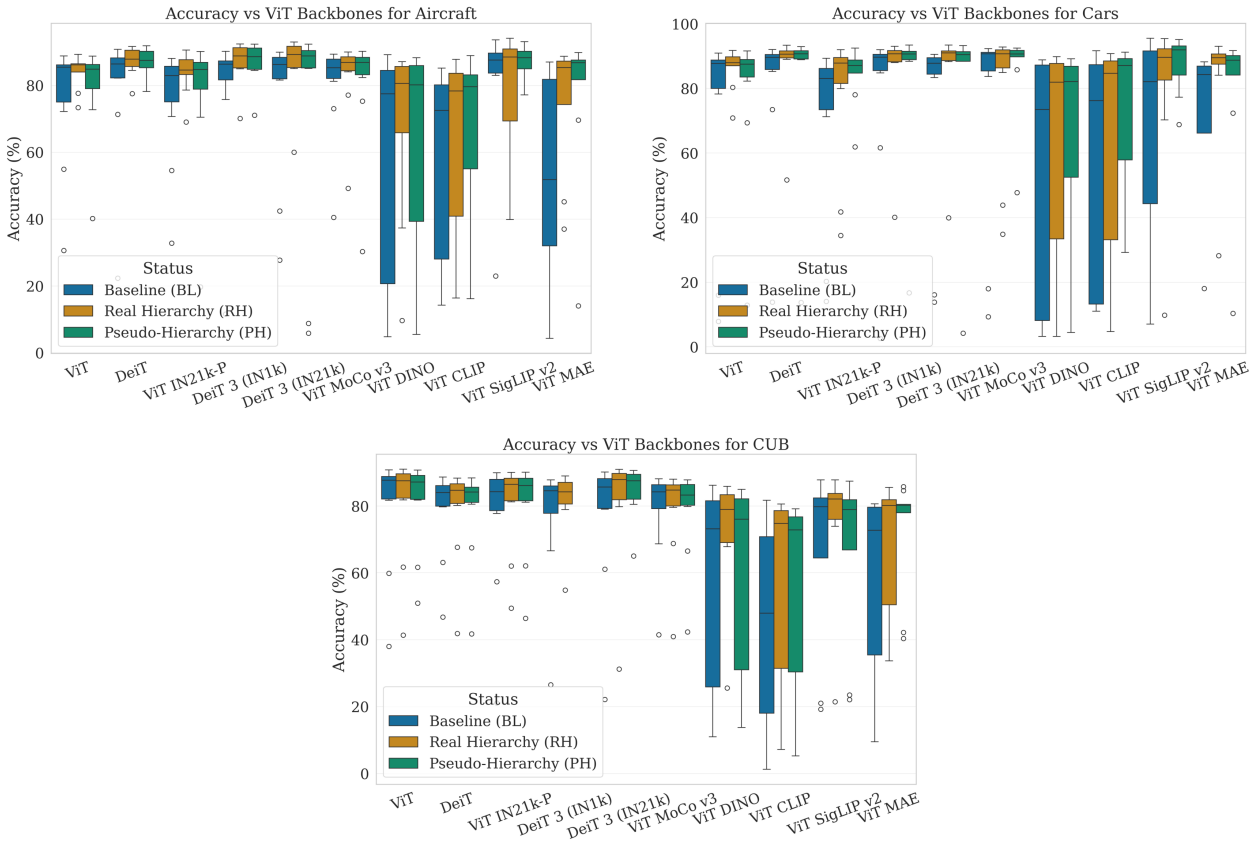


Figure 2. Performance of different ViT models across 3 settings: **Baselines** (Without hierarchy), **Real Hierarchies** and **Pseudo-Hierarchy** and 3 datasets. Pseudo-hierarchy for Aircraft and Cars use 50% cluster ratio, while CUB uses 70%.

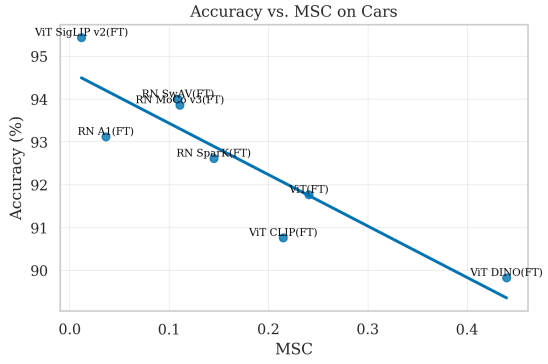


Figure 3. Plot showing Accuracy vs. MSC for the Top-2 and Worst-2 ResNet and ViT fine-tuned models. ViT SigLiP is far away from the rest of the ViT.

5.3. Pseudo-hierarchy on FGIR

5.3.1. Robustness improvements over Baseline

Our pseudo-hierarchy method generally improves robustness of hyperparameter settings over baseline, as shown in Fig.1 and Fig.2. The largest decreases in variance are observed for ViT backbones, such as MAE and SigLipv2, showing that pseudo-hierarchical clustering can stabilize predictions for models that are otherwise sensitive to hyperparameter choices. Across most models, the robustness gains from pseudo-hierarchies are comparable to those achieved with manually defined real hierarchies, and for some models, such as SigLipv2, pseudo-hierarchies even surpass the real hierarchy. These results demonstrate that pseudo-hierarchy offers a lightweight alternative to real hierarchical labels, improving model robustness across hyperparameters without the hefty cost of manual annotation.

5.3.2. Pseudo-hierarchy impact on accuracy

While pseudo-hierarchies generally improve robustness, their impact on mean accuracy is modest for most models. Across most of backbones, accuracy gains range from 1% to 5%, indicating that pseudo-hierarchy primarily stabilizes predictions rather than dramatically increasing overall performance. However, certain ViT backbones show substantial improvements: SigLipv2 gains up to 31.2% accuracy increase, CLIP gains 34.3%, and ViT MAE gains 51.8%. Overall, these results show that while pseudo-hierarchies improve robustness, the corresponding gains in accuracy are

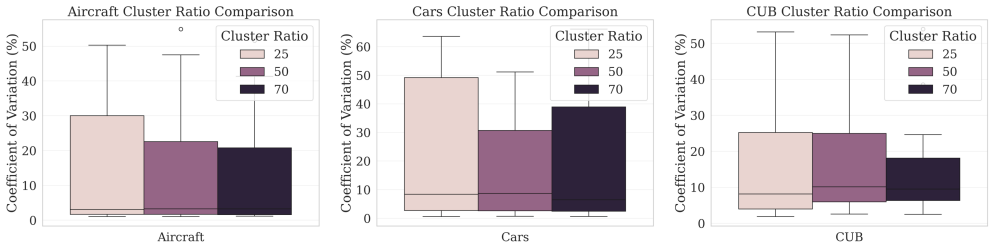


Figure 4. Coefficient of Variations (CV) across 3 different cluster ratio: 25%, 50%, and 70% and 3 datasets. Lower is better.

smaller, barring a few exceptions.

5.3.3. Impact on dataset group

The impact of pseudo-hierarchies varies across datasets. On the machine-centric datasets, **FGVC Aircraft** and **Stanford Cars**, pseudo-hierarchy generally improves robustness compared to the baseline. Most backbones exhibit improved robustness across hyperparameter settings, indicating more stable training behavior. Only a few exceptions are observed among self-supervised models such as SwAV and SupCon. In contrast, the results on **CUB-200-2011** are more mixed. While some models still benefit from pseudo-hierarchies, many show robustness similar to the baseline, and a few backbones obtain worse robustness. Overall, the robustness improvements on CUB are smaller than those observed on Aircraft and Cars. This trend may reflect dataset-specific characteristics discussed in our backbone analysis 5.1.2 and tied to section 5.2.2, where more static shapes in Aircraft and Cars compared to more dynamic shapes in CUB, could influence the effectiveness of hierarchy in improving robustness.

6. Ablation Studies

6.1. Ablation on Cluster Ratios

We evaluate three pseudo-hierarchy cluster ratios (25%, 50%, and 70%) to analyze the impact of the number of coarse clusters on model robustness and performance. These results are shown in Fig.4. Across all datasets, the 25% ratio consistently obtains the highest CV. In contrast, ratios of 50% and 70% generally produce more stable results. For the Aircraft and CUB datasets, the 70% ratio yields the lowest CV, while the Cars dataset yields the lowest CV with the 50% ratio.

The inferior performance of lower cluster ratios may be attributed to the number of fine-grained classes in each dataset. Smaller ratios produce a relatively small number of coarse clusters. This can lead to overly broad groupings that dilute semantic similarity within clusters, reducing the effectiveness of the pseudo-hierarchy. Overall, these results indicate that higher cluster ratios (50%–70%) provide better pseudo-hierarchies, while small ratios, such as 25% tend to degrade robustness.

6.2. Ablation on Other Clustering Algorithm

We compare our proposed pseudo-hierarchy construction with a commonly used clustering baseline, K-Means, on Aircraft with ViT Deit3 backbone as shown in Fig.6. Our method consistently obtains higher accuracy than K-Means, indicating improved robustness while also obtaining better accuracy. This result highlights the importance of the clustering strategy used to construct pseudo-hierarchies. Suboptimal cluster assignments may degrade both performance and robustness, while our proposed bipartite matching produces more stable hierarchical groupings, leading to improved overall performance of the hierarchical classifier.

6.3. Ablation on Other Datasets

To evaluate whether our approach generalizes beyond the primary dataset, we conduct experiments on the SoyLocal dataset. SoyLocal is an Ultra Fine-Grained Image Recognition (UFGIR) dataset consisting of high-resolution soybean leaf images, containing 600 images across 200 fine-grained classes. Due to the extremely small number of samples per class, the dataset presents a challenging setting of unstable training. Fig.7 shows the results of our pseudo-hierarchy approach across different clustering ratios (10, 20, 30, 40, 50, 70, and 90%). Across all ratios, the pseudo-hierarchy consistently achieves higher classification mean accuracy and lower variance compared to the baseline.

6.4. Ablation on Backbone Size

We analyze the impact of backbone capacity by evaluating three model sizes from the DeiT3 family: Base, Large, and Huge. As shown in Fig. 8, increasing the backbone size generally improves robustness under pseudo-hierarchical supervision. The DeiT3-Large model achieves a modest reduction in the Coefficient of Variation (CV) compared to DeiT3-Base, while DeiT3-Huge produces a substantially larger reduction in CV. This trend indicates that larger models benefit more strongly from the pseudo-hierarchical structure, likely due to their increased representational capacity. Overall, these results suggest that pseudo-hierarchy becomes increasingly effective as backbone capacity grows, providing stronger robustness improvements for larger models.

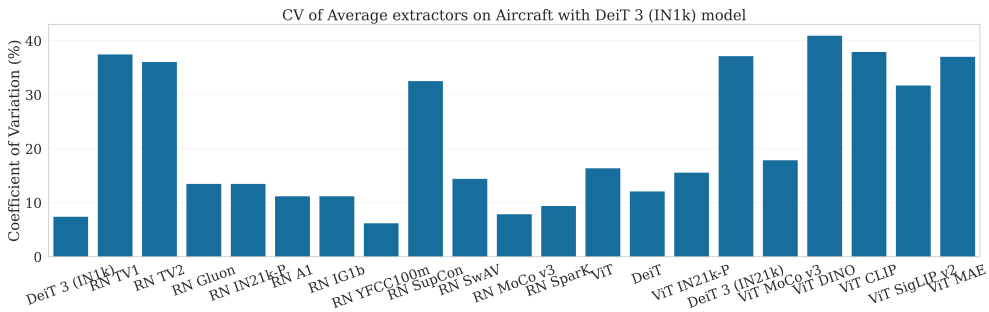


Figure 5. Comparison of 21 feature extractor on Deit3 backbone on Aircraft measured using Coefficient of Variation (CV). Lower is better.

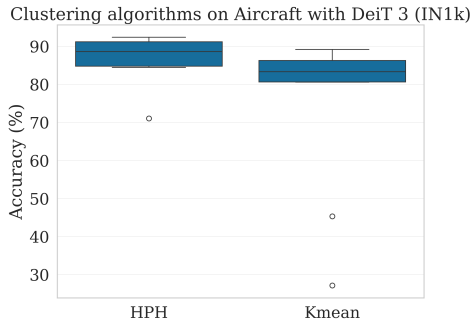


Figure 6. Comparison of Performance between our algorithm: Hungarian Pseudo-Hierarchy (HPH) vs other clustering algorithm.

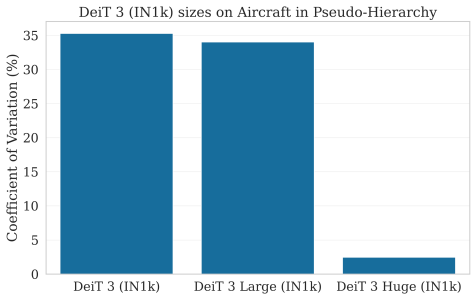


Figure 8. Comparison of Deit3 backbone sizes on Aircraft measured using Coefficient of Variation (CV). Lower is better.

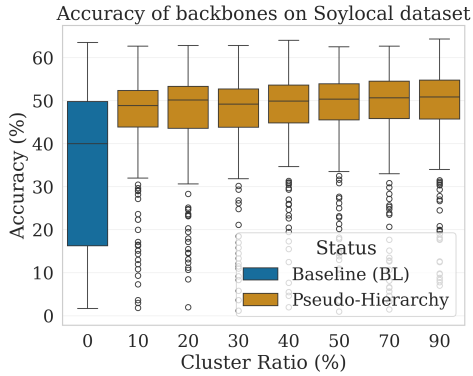


Figure 7. Comparison of 21 backbones on Soylocal dataset. Our method consistently obtains higher accuracy.

6.5. Ablation on Different Extractors

We further analyze the impact of feature extractor choice when constructing pseudo-hierarchies. Using DeiT3 as the training backbone on the Aircraft dataset, we evaluated 21 different pretrained backbones as feature extractors for clustering. The results are shown in Fig. 5. Among the evaluated extractors, RN YFCC100M and the training backbone itself, DeiT3 (IN1K), produce some of the lowest CV, indicating stronger robustness compared to other feature extractors. These results suggest that strong pretrained representations are important for reliable pseudo-hierarchy construction. In practice, using the same backbone for feature extraction and training can serve as a simple and effective choice without requiring additional specialized extractors.

7. Conclusion

The choice of backbone plays a critical role in FGIR; certain backbones may be perceived as inferior due to hyperparameter choices, but incorporating hierarchical supervision into the training of FGIR models can improve robustness to optimization hyperparameters and therefore their overall performance. Furthermore, employing pseudo-hierarchies can be an effective alternative to improve hyperparameter robustness without requiring a rigid labeling scheme.

Pseudo-hierarchies consistently reduce performance variance across training configurations, leading to more stable optimization, and reducing training costs. We observe that these benefits are more pronounced for machine-centric datasets and larger ViT backbones, suggesting that dataset structure and representation capacity influence the effectiveness of hierarchical supervision.

Overall, our findings indicate that pseudo-hierarchical supervision provides an efficient approach for improving robustness in FGIR systems, offering a practical alternative when hierarchical labels are unavailable.

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